

Packet Tracer – Basic BGP Configuration Between Two Autonomous Systems

Objectives

Part 1: Build a Multi-AS Topology

Part 2: Configure IP Addressing

Part 3: Configure BGP Between Two Autonomous Systems

Part 4: Advertise Networks Using BGP

Part 5: Verify BGP Neighbor Relationships

Part 6: Test End-to-End Connectivity

Background / Scenario

You are the network engineer responsible for connecting two organizations to each other.

Each organization owns its own Autonomous System (AS) and must exchange routing information using the **Border Gateway Protocol (BGP)**.

In this lab, you will configure:

- Two routers representing different Autonomous Systems
- BGP peering between the routers
- Network advertisement using BGP
- Route verification
- End-to-end connectivity testing

This lab introduces the fundamentals of Internet routing and demonstrates how BGP allows independent networks to exchange reachability information.

Instructions

Part 1: Build the Topology

Step 1: Add Devices

a. Open **Cisco Packet Tracer**.

b. Add:

- 2 Routers (2911 recommended)
- 2 PCs

Rename:

- Router0 → **R1-AS65001**
 - Router1 → **R2-AS65002**
 - PC0 → **PC-A**
 - PC1 → **PC-B**
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Step 2: Connect Devices

Use Copper Straight-Through cables.

Connect:

Device Interface		Device Interface	
PC-A	FastEthernet0	R1	GigabitEthernet0/0
PC-B	FastEthernet0	R2	GigabitEthernet0/0
R1	GigabitEthernet0/1	R2	GigabitEthernet0/1

Wait until interfaces initialize.

Part 2: Configure IP Addressing

Network Plan

Device Interface IP Address		
PC-A	NIC	192.168.10.10/24
R1	G0/0	192.168.10.1/24
R1	G0/1	10.0.12.1/30

Device Interface IP Address

R2	G0/1	10.0.12.2/30
R2	G0/0	192.168.20.1/24
PC-B	NIC	192.168.20.10/24

Step 1: Configure R1

```
enable
configure terminal

interface g0/0
ip address 192.168.10.1 255.255.255.0
no shutdown

interface g0/1
ip address 10.0.12.1 255.255.255.252
no shutdown

end
```

Step 2: Configure R2

```
enable
configure terminal

interface g0/0
ip address 192.168.20.1 255.255.255.0
no shutdown

interface g0/1
ip address 10.0.12.2 255.255.255.252
no shutdown

end
```

Step 3: Configure PC-A

Desktop → IP Configuration

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Step 4: Configure PC-B

Desktop → IP Configuration

IP Address: 192.168.20.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

Part 3: Configure BGP

Understanding the Design

AS Numbers:

R1 = AS 65001

R2 = AS 65002

BGP Neighbors:

R1 <-----> R2

10.0.12.1 10.0.12.2

Step 1: Configure BGP on R1

enable

configure terminal

router bgp 65001

neighbor 10.0.12.2 remote-as 65002

network 192.168.10.0 mask 255.255.255.0

end

Step 2: Configure BGP on R2

enable

configure terminal

router bgp 65002

neighbor 10.0.12.1 remote-as 65001

network 192.168.20.0 mask 255.255.255.0

end

What Do the Commands Mean?

Configure Local AS

router bgp 65001

Starts the BGP process and places the router in AS 65001.

Configure Neighbor

neighbor 10.0.12.2 remote-as 65002

Tells R1:

"My BGP neighbor is 10.0.12.2 and it belongs to AS 65002."

Advertise a Network

network 192.168.10.0 mask 255.255.255.0

Tells BGP:

"Advertise this network to my BGP neighbors."

Part 4: Verify BGP Operation

Step 1: Verify Neighbor Relationship

On R1:

```
show ip bgp summary
```

Expected:

Neighbor	State/PfxRcd
10.0.12.2	Established

On R2:

```
show ip bgp summary
```

Expected:

Neighbor	State/PfxRcd
10.0.12.1	Established

Step 2: Verify Learned Routes

On R1:

```
show ip route
```

Expected:

```
B 192.168.20.0/24
```

On R2:

```
show ip route
```

Expected:

```
B 192.168.10.0/24
```

The **B** indicates a route learned through BGP.

Step 3: View BGP Table

show ip bgp

Expected output should include:

192.168.10.0

192.168.20.0

Part 5: Test Connectivity

Step 1: Ping Between Routers

From R1:

ping 10.0.12.2

Success expected.

From R2:

ping 10.0.12.1

Success expected.

Step 2: Ping Across Autonomous Systems

From PC-A:

ping 192.168.20.10

Expected:

Reply from 192.168.20.10

4 successful replies.

From PC-B:

ping 192.168.10.10

Expected:

Reply from 192.168.10.10

4 successful replies.

Troubleshooting Tips

If BGP Neighbor State is not Established:

✓ Verify both routers can ping each other

ping 10.0.12.2

✓ Verify correct AS numbers

show running-config

✓ Verify neighbor IP addresses

✓ Ensure interfaces are up

show ip interface brief

✓ Verify the network command exactly matches the route in the routing table

What You Learned

- How BGP connects different Autonomous Systems
 - How eBGP neighbors are configured
 - How networks are advertised using BGP
 - How BGP routes appear in the routing table
 - How Internet-style routing works
 - How to verify BGP neighbor relationships
 - How BGP exchanges reachability information between organizations
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Challenge Extensions (Optional)

Challenge 1: Add a Third Autonomous System

Add a third router:

R3 = AS 65003

Configure:

R1 <-> R2 <-> R3

Verify route propagation between all ASs.

Challenge 2: Advertise Multiple Networks

Add a second LAN behind each router.

Example:

192.168.11.0/24

192.168.21.0/24

Advertise both using BGP.

Verify routes appear in neighbor BGP tables.

Challenge 3: Observe BGP in Simulation Mode

Switch Packet Tracer to:

Simulation Mode

Generate pings and observe:

- TCP Port 179 sessions
 - BGP Open messages
 - BGP Update messages
 - Route advertisements
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Challenge 4: Break the BGP Session

Shutdown the inter-router interface:

interface g0/1

shutdown

Observe:

show ip bgp summary

Notice the neighbor relationship goes down.

Restore connectivity and verify BGP reconverges.